

# One-Way ANOVA — Part B

## Within & Between Group Variability Student Worksheet

Name: \_\_\_\_\_

Date: April \_\_\_\_, 2026

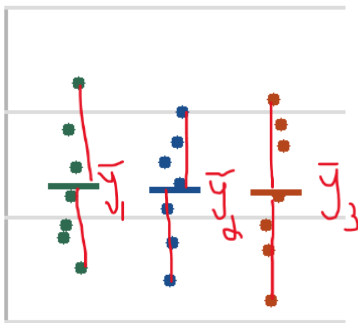
**Context:** This section builds conceptual understanding of the two sources of variability that one-way ANOVA uses to compare group means: **within-group variability** (noise) and **between-group variability** (signal). The response variable ( $y$ ) is plotted vertically in all dot-plots.

### 1 Within-Group Variability

**Within-group variability** measures how spread-out individual response variable values are around their own group mean  $\bar{y}_i$ . It is the **noise** in ANOVA — variation that exists even within groups receiving the same treatment.

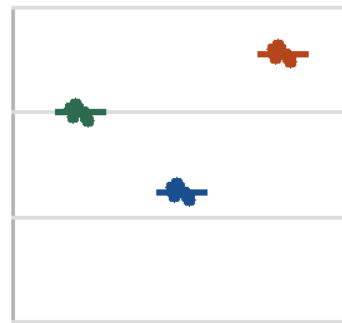
The dot-plots below show the response variable ( $y$  = exam score) on the **vertical axis** and groups (A, B, C) on the horizontal axis.

High Within-Group Variability



*Dots spread far within each group*

Low Within-Group Variability



*Dots tight within each group*

#### 1a Fill in the blanks

When within-group variability is **large**, individual response values are spread **far** from their group mean  $\bar{y}_i$ .

This makes it **HARD** for ANOVA to detect real differences between groups, because

the **noise** grows larger.

#### 1b Identify which plot shows larger within-group variability and explain why it matters

**The left plot has the higher within group variability because the points dramatically deviate from each groups mean. When the within group variability spread is large, the one-way ANOVA cannot detect the real differences between the groups means.**

## 2 Between-Group Variability

**Between-group variability** measures how far apart the **group means  $\bar{y}_i$**  are from the overall mean  **$\bar{\bar{y}}$**  (the grand mean across all observations). It is the **signal** — evidence that the groups actually differ.

Using the data from the study: group means are  $\bar{y}_A = 72$ ,  $\bar{y}_B = 85$ ,  $\bar{y}_C = 91$ , and the overall mean is  $\bar{\bar{y}} = 83$ .

### 2a Calculate deviations

| Group   | Group Mean $\bar{y}_i$ | Overall Mean $\bar{\bar{y}}$ | Deviation $\bar{y}_i - \bar{\bar{y}}$ |
|---------|------------------------|------------------------------|---------------------------------------|
| Group A | 72                     | 83                           | $72 - 83 = -11$ points                |
| Group B | 85                     | 83                           | $85 - 83 = 2$ points                  |
| Group C | 91                     | 83                           | $91 - 83 = 8$ points                  |

### 2b Interpretation

The group with the largest deviation from the overall mean is contributing most to between-group variability. Which group is it, and what does this suggest about the effectiveness of that study method?

**Group A is deviating the most from the overall mean with 11 point below and group C is 8 points above. The lowest performing group is re-reading.**

### 3 The F-Ratio: Signal Over Noise

The **F-statistic** is computed as:

$$F = \frac{\text{Between-Group Variability}}{\text{Within-Group Variability}} = \frac{\text{MSB}}{\text{MSW}}$$

$$\frac{\text{Mean Square Between}}{\text{Mean Square Within}}$$

$$\frac{\text{Squared Deviations on average}}{\text{Squared Deviations on average}}$$

#### 3a Fill in the blanks

When  $F \gg 1$ , the between-group signal is **large** relative to within-group noise. This **supports** the alternative hypothesis that at least one group mean  $\bar{y}_i$  differs from the others.

When  $F \approx 1$ , the between-group signal is **similar in size** to within-group noise. This is consistent

with **the null hypothesis that the populations means are the same.**

**Can F be negative? No, Reason: Since we are square the deviations, the F statistic has to be 0 or more.**

**3b Interpret  $F = 9.14$  (from the ANOVA output table in Part A) means that the between group variability is approximately 9 times larger than within group variability. The strong signal relative to the noise is apparen providing evidence against the null hypothesis.**

### 4 Equal Variance Condition

ANOVA assumes that within-group variability is **roughly equal across all groups** — this is called **homoscedasticity**. If violated, the pooled MSW (denominator of F) is distorted.

#### 4a Check the condition

| Group | Scores             | Std Dev (approx.) |
|-------|--------------------|-------------------|
| A     | 78, 80, 79, 81, 77 | ~1.6              |
| B     | 42, 65, 88, 91, 55 | ~21.1             |
| C     | 72, 74, 73, 75, 71 | ~1.6              |

Ratio of largest to smallest SD:  $21.1/1.6 \approx 13$  Is the condition met (rule: largest SD  $\leq 2 \times$

smallest)? **No, the assumption has been clearly violated**

Explain what this violation means for interpreting the F-statistic. **-**

